



The alarm limits may set to any value between **0.0** and **99.9%**, where the value entered relates to a **percentage deviation** from each oxygen setpoint in the firing range.

If the burner is firing, it is only possible to make an adjustment to the option parameters that relate to the selected profile. If the burner is not firing, the alarm values for any profile selection may be adjusted.

An oxygen low alarm will be caused by one or more of the following events:

- The level drops below 0.5% O₂ for 30 seconds
- The level drops below the low alarm limit for 2 minutes
- The level drops below **twice** the low alarm limit for 30 seconds

An oxygen high alarm will be caused by one or more of the following events:

- The level rises above the high alarm limit for 2 minutes
- The level rises above **twice** the high alarm limit for 30 seconds

When 2 probes the low alarm will also be caused by the following event:

- The difference between the oxygen values exceeds the value set in option parameter 42.4 for 2 minutes

Option 42.0 – Second Oxygen probe interface serial number LV3

Each CANbus device connected to the system has a unique serial number allocated to it during production testing. This number cannot be changed and is used to uniquely identify each device.

When the system is in commissioning mode the serial number of all the connected devices is read by the PPC6000 and displayed as a 'list' to allow the selection of the relevant serial number unit to be made.

Scrolling the list in an upward direction will scroll through all oxygen probe interface options available until '0, nothing' is displayed. Enter this value to disconnect the specified device from any hardware.

The FIREYE oxygen probe and associated electronics are not fail-safe. It is for this reason that the maximum amount of oxygen trim allowed must be limited by the commissioning engineer using option parameters 32.1 to 32.4. The trim must be limited to ensure that in the event of a probe fault (incorrect O₂ level), the combustion does not become unsafe.

For applications demanding large amounts of oxygen trim to be applied, or where oxygen monitoring is especially critical, a second oxygen probe and controller can be selected here. The two oxygen readings from the probes can be compared when the burner is running to provide fail-safe oxygen monitoring.

Option 42.1 – Second Oxygen probe calibration offset value (0 – 999) LV3

This option parameter is only available if option 31.0 is non-zero.

This option parameter must be set to the calibration offset value that is specified with the probe supplied.

Option 42.2 – Second Oxygen probe calibration gain value (0 – 999) LV3



This option parameter is only available if option 31.0 is non-zero.

This option parameter must be set to the calibration gain value that is specified with the probe supplied.

Option 42.3 – Second Oxygen probe calibrate enable (0 – 2) LV3

This option parameter is only available if option 31.0 is non-zero.

This option parameter must be used when calibrating the second FIREYE oxygen probe. There are three possible values, as seen on the following chart:

Option parameter 31.3 value	Meaning
0	No calibrate. This is the normal operating condition, where the probe is used to measure the oxygen concentration in the flue.
1	Calibrate in air. The probe will be calibrated for 'offset' at the normal atmospheric oxygen concentration.
2	Calibrate in reference gas. The probe will be calibrated for 'gain' in a reference gas with a nominal oxygen concentration the level specified in option parameter 30.3 (typically 3.0%).

Once this calibration has been successfully completed the values in 31.1 and 31.2 will be automatically updated.

Option 42.4 – Max oxygen variation (0 – 9.9 %) LV3

Where the second oxygen probe is being used as a cross-check on the first one, the maximum permitted variation between the two oxygen readings should be entered here. It is suggested that a value of around 0.5 % be used, but the application may demand a higher setting if the two probes are not in exactly the same part of the flue. The control will lockout within 30 seconds of the two oxygen readings being outside the specified window.

When this parameter is non-zero, the flue oxygen limits will be checked on a 'worst case' basis. This means that the low oxygen limit (see options 38.X, 39.X) is checked against the lower of the two oxygen probe readings, and the high oxygen limit (see options 40.X, 41.X) is checked against the higher of the two oxygen probe readings. If there is a fault with second oxygen probe, the burner will also lockout.

If this parameter is left at zero (0.0) no checking will be done, but the oxygen and temperatures associated with second oxygen probe will still be available on the engineers key (EK75-79).

Note: When fail-safe oxygen monitoring is employed using two probes, each probe must be tested and calibrated at intervals not less frequent than once every six months.



Option 42.5 – Max flue temp variation (0 to 999 °C) LV3

If non-zero, the burner will lockout if the two oxygen probes vary by more than this many degrees (Celsius). This can be used in applications where flue temperature is especially critical.

Option 42.7 – Air pressure monitoring* – input number LV3

Available on firmware version 1.3 onwards (EK56).

Option 42.8 – Air pressure monitoring* – Span LV3

Option 42.9 – Air pressure monitoring* – max error LV3

* For options 42.7, 42.8, 42.9 - EK98 shows air pressure error if in RUN and MODULATION only. If not modulating or in a commission mode, it shows the actual pressure value otherwise it can't be seen on the NX610 display in commission mode. Air pressure will show as a display value on the NX610 display and on the touchscreen if updated to 1.039 or later. Currently air pressure setpoints cannot be seen in the profile tables in comfire or on the touchscreen.

Option 43.0 – Water level / expansion unit LV3 - Reserved**

Option 43.1 – Water level / expansion unit LV3 - Reserved**

* For options 43.0 and 43.1 - Firmware 1.039 or later.

Option 44.0 - Set real time clock (0/1) LV1

Setting this option 1 enables the user to set the clock in the control used for fault logging. First set this option to 1 then adjust the options 44.1 to 44.6. When these are correct, check that this option is still at 1, then press the RUN key then the ENTER key.

Option 44.1 – Set Year (00 – 99) LV1

See "SET CLOCK" Example after Option 44.7

Option 44.2 – Set Month (1 – 12) LV1

See "SET CLOCK" Example after Option 44.7

Option 44.3 – Set Day of Month (1 – 31) LV1

See "SET CLOCK" Example after Option 44.7



Option 44.4 – Set Day of Week (1 - 7) LV1

For reference 1=Sunday, 2=Monday through 7=Saturday
See “SET CLOCK” Example after Option 44.7

Option 44.5 – Set Hours (0 - 23) LV1

Hours use 24 hour format. AM= 0-11, PM=12-23; For reference 0=midnight, 12=noon, 18= 6 pm (12+6)
See “SET CLOCK” Example after Option 44.7

Option 44.6 – Set Minutes (0 - 59) LV1

See “SET CLOCK” Example after Option 44.7

Option 44.7 – Set Seconds (0 - 59) LV1

See “SET CLOCK” Example after Option 44.7

“SET CLOCK” EXAMPLE

To Set 6:30:45, PM, Tuesday, December, 25, 2007

The Values should be entered as follows:

44.1 = 7	(Year)
44.2 = 12	(Month)
44.3 = 25	(Day of Month)
44.4 = 3	(Day of Week)
44.5 = 18	(Hours) (12+6)
44.6 = 30	(Minutes)
44.7 = 45	(Seconds)

Option 44.9 – Reset Fault Log (0 - 1) LV3

This option clears the current the fault history.

Set the option parameter to 1, then press RUN, than ENTER. The control will erase the fault history and return to run mode.

Option 45.0 - Erase / Restore enable (0 - 1) LV3



CAUTION

- If this value is displayed as a '2', the option parameter data has been 'up-loaded' into the control, ensure all option parameters are set to match the requirements of the burner to which it is connected. Failure to do so could cause a hazardous condition to occur.
- After all option parameters have been checked to match the requirements of the burner to which it is connected, reset this option parameter to zero to allow the system to operate.
- If an attempt is made to operate the system with this option parameter set to '2' a fault will be generated and the control will perform a non-volatile lockout, preventing the burner from firing.

In order to erase information in memory for a specific profile selection (see option parameter 45.1), this option must be set to '1'. Once the control has left set-up mode, this parameter will automatically be reset to '0'. In order to restore information to the control from the back-up held in the display (see option parameter 45.2), this option must be set to a '1'. Once the control has left set-up mode, this parameter will automatically be reset to '0'.

If this option parameter is set to '2' see caution message above.

Option 45.1 - Erase command (0 - 5) LV3

When erase enable (option 45.0) is set to '1', this option parameter determines which information will be erased when the control leaves option set mode.

- 0 - No erase.
- 1 - Erase profile 1.
- 2 - Erase profile 2.
- 3 - Erase profile 3.
- 4 - Erase profile 4.
- 5 - Erase all setpoints and initialize all option parameters.

If 5 is selected, **all profile setpoints will be erased and all option parameters will be set to default values.** The control will be initialized to the same state as when it was shipped.

NOTE: You must exit commissioning mode to begin the erase.

Option 45.2 - Restore command (0 - 999) LV3

MEMORY TEST: To determine if the NX610 keypad display contains a valid memory, press both left and right cursor (arrow) keys simultaneously. The display will change to LOCAL MODE and toggle between two information screens. One of these will contain the capitol letters FL CDBKP. If all " _ _ _ _ " appear, there is no valid memory in the display. If no valid option parameter and profile is available from previous downloads using Comfire 2 software, all data and combustion curves will need to be entered by hand. Please refer to Fireeye bulletin NEX-1502 (Comfire 2), available at www.Fireeye.com. Comfire 2 will not restore the servomotor serial numbers which will need to be hand entered in Option 3.0 to 3.9 **BEFORE** restarting the PPC6000. A valid backup is achieved after the burner has been running in "AUTO" for approximately 20 minutes.



When erase / restore enable (option 45.0) is set to '1', this option parameter allows the back-up data held in the display to be loaded into the PPC6000, this may take up to 5 minutes to complete during which time the control will not allow the burner to operate.

IMPORTANT: In order for the restore command to function, the new control must have the same communications address (Option Parameter 0.2) as the original control. That is, if option 0.2 was set to 2 (boiler #2 for example), the commissioning engineer must set options 0.2 of the new control to 2 **BEFORE** setting option 45.2 to 100. The default communications address of the control is 0. If multiple PPC6000's have been connected together for sequencing, it is likely that the communications address has been changed to match the boiler number.

To restore the data from the back-up held in the NX610 or the NXTSD104 display, set this option parameter to '100'.

If **100** is selected, **the data from the back-up data held in the display will be loaded into the PPC6000.**

A back-up of the data held in the PPC6000 is automatically stored in the display when the control is in modulation status.

NOTE: You must exit commissioning mode to initiate the restore.

5.6 Commission ratio mode

5.6.1 Description

Commission ratio mode is used for entering a new profile or changing an existing profile. Using commission ratio mode, it is possible to step through each setpoint including close, purge and ignition. Once a setpoint has been entered, it is not possible to go back and modify it again in the same commissioning session. To do this, adjust ratio mode must be used after all the required setpoints have been entered.

5.6.2 Using commission ratio mode



CAUTION

- Use extreme care while commissioning the system. While operating in commissioning mode the safety of the system is the sole responsibility of the commissioning engineer.
- Ensure a pre-purge position is entered for the relevant drives.
- Incorrect positioning of either fuel or air could cause a hazardous situation to occur.
- If the servo-motor positions have been up-loaded it is essential that the combustion is verified at each firing position to ensure a hazardous condition cannot occur, and for the data to be stored in the control.
- It is recommended that the close position for each servo motor is re-entered as part of the burner service regime, to compensate for wear in the servo-motor micro-switch during operation.
- Once all profile positions have been adjusted/entered it is essential that all profile points are checked to verify that the fuel air ratio is acceptable for the appliance being controlled.
- To adjust the position of a servo motor or inverter, select the motor using the scroll keys, then use the UP/DOWN keys to change the value.



Each setpoint in the profile is referred to with a letter and number (refer to the table below).

No.	Name	Comments
P0	Close	This defines the positions the motors will move to when the burner is off, and must be set to the lowest position each motor can reach.
P1	Purge	This defines the positions the motor will move to when the burner is purging, and can be set as required for each servo-motor.
P2	Ignition	Position during ignition. May be outside the normal firing range.
P3	Low fire	Low fire setpoint, (unless limited modulation range is selected).
.....		
Px	High fire	High fire setpoint, (unless limited modulation range is selected), which is the last setpoint entered (max. P24)

The procedure for entering or modifying a profile is as follows :

1. Enter commission ratio mode (see section 5.3.1). P0 (close) is displayed.
2. If the close setpoint has recently been entered and no change is required, press **NEXT**. If servicing the burner or the position has not been entered before, move each motor to its lowest position and press **ENTER**. P1 (purge) is displayed (flashing).
3. Select the burner and the required profile. If the purge position has been set before the selected motors will automatically move to their relevant P1 positions and stop, if no change is required press **NEXT**. Otherwise, move each motor to its required pre-purge position and press **ENTER**. If the purge setpoint has not been entered before the servo-motors will remain at their respective closed (P0) positions until each motor has been moved to its required position. Once all selected motors are at the correct position press **ENTER**. P1 (purge) is displayed. Please note that if a purge position for a particular drive has not been entered before the drive will remain at its closed position. **The engineer must ensure that the main combustion air damper is set to open enough to purge the boiler.**
4. Wait for the pre-purge time to elapse. If the ignition setpoint has been entered before, the drives will move to their ignition positions once all drives stop moving P2 (ignition) will be displayed, if a change is required or the ignition setpoint has not been entered before, move all the motors being used to their required ignition positions and press **ENTER**.
5. **To attempt ignition of the burner, hold down the NEXT key for approximately three seconds.** If it is wished to change the ignition position adjust the relevant drive(s) and press **ENTER**, if the position is acceptable press **NEXT**. **Once the relevant safety times have elapsed P3 will be displayed.**
6. If the low fire setpoint has been entered before and no change is required, press **NEXT**. Otherwise, move each drive to the required low fire position and press **ENTER**. P4 (next profile setpoint above low fire) is displayed.
7. Repeat step 6 for each required profile setpoint, up to a minimum of P4 and a maximum of P23.
8. Leave commission ratio mode. The last profile setpoint entered will become the high fire setpoint.
 - If a controlled shutdown occurs, the controller will return to step 2. The setpoints entered in the current commissioning session are not lost and the **NEXT** key may be used to step through the start-up sequence and fire the burner.
 - If a non-volatile lockout occurs, the setpoints are kept in the same way as for a controlled shutdown. It will be necessary to remove all faults before moving further than step 2.
 - If power is removed from the controller the setpoints entered in the current commissioning session will be lost.

5.6.3 Leaving commission ratio mode

Commission ratio mode →   → Run mode

To leave commission ratio mode and return to run mode, press the key labeled **RUN** followed by the key labeled **ENTER**.

- **If P3 (or higher) has been commissioned, the existing profile will be overwritten.** If you do not wish the existing profile to be overwritten, disconnect the power to the control without pressing **RUN** and **ENTER**.
- **Only the setpoints used in the current commissioning session will be stored.** For example, if an existing profile has setpoints up to P15 but only the setpoints up to P10 were viewed or altered, then only the setpoints up to P10 will be stored. Therefore, it is *vital* that **NEXT** is pressed to get to the last setpoint in the profile before leaving commission ratio mode. This does not apply if only the close, purge or ignition setpoints (P0, P1 or P2) are altered.

5.7 Adjust ratio mode

5.7.1 Description

Adjust ratio mode is used only for changing an existing profile.

- Using adjust ratio mode, it is possible to modify the motor positions for any setpoint in the firing range from low to high fire (P3 and higher).
- The setpoints may be chosen in any order.
- It is **not** possible to modify any motor position so that it is higher than the high fire setpoint.
- It is **not** possible to modify the close, purge or ignition positions (P0, P1 or P2 respectively). For this, commission ratio mode must be used.

5.7.2 Entering adjust ratio mode

Run mode, burner on →  →   → Supplier passcode → 

To enter adjust ratio mode, turn the burner on. Once the burner begun its startup sequence, press the key labeled **COM** and obtain the adjust ratio passcode value using the **UP/DOWN** keys. Once the correct passcode value has been obtained, press the key labeled **ENTER**.

If the entered passcode was correct, 'A n' will be seen on the display, where n is the number of the setpoint currently under adjustment.

Entering the commission ration passcode wile the burner is **ON** enters "Adjust Ratio" mode

5.7.3 Using adjust ratio mode



CAUTION

- Use extreme care while commissioning the system. While operating in adjust ratio mode the safety of the system is the sole responsibility of the commissioning engineer.
- Incorrect positioning of any drive could cause a hazardous situation to occur.

- To adjust the position of a servo-motor or inverter, select the relevant drive and use the **UP/DOWN** keys.
- To change the setpoint being modified, using the scroll key select the setpoint and then use the **UP/DOWN** keys.

Each setpoint in the profile is referred to with a letter and number.

No.	Name	Comments
A0	Close	The close setpoint cannot be altered in adjust ratio mode.
A1	Purge	The purge setpoint cannot be altered in adjust ratio mode.
A2	Ignition	The ignition setpoint cannot be altered in adjust ratio mode.
A3	Low fire	Low fire setpoint.
.....		
Ax	High fire	High fire setpoint, which was the last setpoint entered (max P24)

The procedure for modifying a setpoint is as follows:

1. Enter adjust ratio mode. 'A n' is displayed in the window, where n is the number of the setpoint that is closest to the current firing position.
2. Use the scroll key to select the setpoint and use the **UP/DOWN** keys to move to the setpoint in the firing range to be altered.
3. Wait for the 'A n' display to stop flashing and modify the drive(s) as required.
4. If the modified positions are not required, press the **NEXT** key to return the drive(s) to their original positions or use the scroll key to move to a different setpoint.
5. Press the **ENTER** key to store the new drive positions for the current setpoint.
6. Repeat steps 2-5 as required. Each time step 5 is completed, the new positions will be stored permanently.
7. Leave adjust ratio mode.

If the 'A n' display is flashing, it will not be possible to adjust the position of any of the drives. This may be for one of the following reasons:

- The burner is not firing. Switch the burner on and wait for the control to begin modulating.
- The current setpoint is A0, A1 or A2. Use commission ratio mode to adjust these setpoints.
- The drive(s) are moving to the required positions. Wait for the drive(s) to stop moving.



5.7.4 Leaving adjust ratio mode

Adjust ratio mode → **RUN** **ENTER** → Run mode

To leave adjust ratio mode and return to run mode, press the key labeled **RUN** followed by the key labeled **ENTER**.

5.8 Interrogating the software issue

There are two methods for finding the software issue:

1. From run mode, press the **COM** key. The Fireeye version number will appear.
2. Use the engineer's key (refer to section 6.6 for details.) and scroll up to EK200 for the current Fireeye operating version.